

Brutal Series - Paper B57

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. An object stands 12 cm in front of a flat (plane) mirror. The mirror is then slid 5 cm straight back, away from the object. By how much does the gap between the object and its image change? [\[Light\]](#)
 - (A) 29 cm
 - (B) 34 cm
 - (C) 10 cm
 - (D) 24 cm
 - (E) 2 cm
 - (F) 5 cm
 - (G) 17 cm

2. A school bag is marked at Rs 1200. A shop first gives a 15% discount on the marked price, and then adds 5% tax on the discounted price. What final amount is paid? [\[Comparing Quantities\]](#)
 - (A) Rs 1080
 - (B) Rs 1000
 - (C) Rs 1020
 - (D) Rs 1050
 - (E) Rs 1260
 - (F) Rs 1140
 - (G) Rs 1071

3. In a percolation test, 200 mL of water completely passes through a soil sample in 25 minutes. What is the percolation rate of this soil in millilitres per minute? [\[Soil\]](#)
 - (A) 16 mL/min
 - (B) 40 mL/min
 - (C) 0.125 mL/min
 - (D) 225 mL/min
 - (E) 5000 mL/min
 - (F) 8 mL/min
 - (G) 12.5 mL/min

4. Add the rational numbers $(-3/4)$ and $(5/6)$. Give the answer in lowest terms. [\[Rational Numbers\]](#)
 - (A) $1/12$
 - (B) $1/2$
 - (C) $2/10$
 - (D) $19/12$
 - (E) $-1/12$
 - (F) $8/10$
 - (G) $-2/12$

5. Two plane mirrors are set facing each other with an angle of 72 degrees between them. A small candle is placed in the gap. How many images of the candle are formed? [\[Light\]](#)
- (A) 72
 - (B) 4
 - (C) 3
 - (D) 360
 - (E) 6
 - (F) 5
 - (G) infinite
6. A toy that cost a shopkeeper Rs 250 is sold for Rs 320. What is the profit percent? [\[Comparing Quantities\]](#)
- (A) 28%
 - (B) 22%
 - (C) 21.875%
 - (D) 35%
 - (E) 25%
 - (F) 32%
 - (G) 70%
7. Soil X lets 250 mL of water pass in 10 minutes, while Soil Y lets the same 250 mL pass in 50 minutes. Which soil is more likely to be sandy, and why? [\[Soil\]](#)
- (A) Soil Y, because water passes through it slower
 - (B) Soil X, because water passes through it much faster
 - (C) Soil X, because it has smaller particles
 - (D) Soil Y, because it drains the fastest
 - (E) Both are equally sandy
 - (F) Soil Y, because it has larger particles
8. Which of these two rational numbers is greater: $-\frac{2}{3}$ or $-\frac{3}{4}$? [\[Rational Numbers\]](#)
- (A) Neither can be compared
 - (B) They are equal
 - (C) $-\frac{3}{4}$ is closer to zero
 - (D) $-\frac{2}{3}$ equals $-\frac{3}{4}$
 - (E) $-\frac{2}{3}$ is greater
 - (F) Both are positive
9. When you look at a wall clock reflected in a plane mirror, the hands appear to read 2:30. What is the real time shown by the clock? [\[Light\]](#)
- (A) 10:30
 - (B) 3:30
 - (C) 9:00
 - (D) 6:30
 - (E) 9:30
 - (F) 2:30

- 10.** Rs 4500 is to be shared between two friends A and B in the ratio 4 : 5. How much MORE money does B get than A? [\[Comparing Quantities\]](#)
- (A) Rs 1000
 - (B) Rs 2000
 - (C) Rs 500
 - (D) Rs 900
 - (E) Rs 2500
 - (F) Rs 450
 - (G) Rs 250
- 11.** Starting from the surface and digging straight down, in what order do you meet the main layers of soil? [\[Soil\]](#)
- (A) Bedrock, then weathered rock, then subsoil, then topsoil
 - (B) Topsoil, then subsoil, then weathered rock, then bedrock
 - (C) Topsoil, then bedrock, then subsoil, then weathered rock
 - (D) Subsoil, then weathered rock, then topsoil, then bedrock
 - (E) Weathered rock, then bedrock, then topsoil, then subsoil
 - (F) Subsoil, then topsoil, then bedrock, then weathered rock
 - (G) Topsoil, then weathered rock, then subsoil, then bedrock
- 12.** Multiply $(-2/5)$ by $(15/-8)$ and give the answer in lowest terms. [\[Rational Numbers\]](#)
- (A) $3/4$
 - (B) $-3/4$
 - (C) $-6/13$
 - (D) $6/13$
 - (E) $3/-4$
 - (F) $30/40$
- 13.** Between two plane mirrors a coin forms exactly 5 images. What is the angle between the mirrors? [\[Light\]](#)
- (A) 5 degrees
 - (B) 30 degrees
 - (C) 90 degrees
 - (D) 72 degrees
 - (E) 36 degrees
 - (F) 45 degrees
 - (G) 60 degrees
- 14.** In a class of 40 children, 60% are girls. Of these girls, 25% wear spectacles. How many girls in the class wear spectacles? [\[Comparing Quantities\]](#)
- (A) 6
 - (B) 24
 - (C) 16
 - (D) 8
 - (E) 4
 - (F) 12
 - (G) 10

15. Three kinds of soil particle are clay, sand and silt. Arrange them in order from the LARGEST particle to the smallest. [\[Soil\]](#)

- (A) Silt, then sand, then clay
- (B) Sand, then silt, then clay
- (C) Sand, then silt, then loam
- (D) Silt, then clay, then sand
- (E) Clay, then silt, then sand
- (F) Sand, then clay, then silt

16. What is the reciprocal (multiplicative inverse) of the rational number $-7/9$? [\[Rational Numbers\]](#)

- (A) $-1 \times 7/9$
- (B) $9/-7$ equals $9/7$
- (C) $-9/7$
- (D) $-7/9$
- (E) $9/7$
- (F) 0
- (G) $7/9$

17. A boy walks straight toward a plane mirror at a steady 2 metres per second. At what speed does his image appear to approach HIM? [\[Light\]](#)

- (A) 2 m/s
- (B) 6 m/s
- (C) 1 m/s
- (D) 3 m/s
- (E) 8 m/s
- (F) 4 m/s
- (G) 0 m/s

18. Find the simple interest earned on Rs 8000 kept at 7.5% per year for 2 years. [\[Comparing Quantities\]](#)

- (A) Rs 480
- (B) Rs 600
- (C) Rs 9200
- (D) Rs 1280
- (E) Rs 1500
- (F) Rs 1200

19. A farmer wants a soil that holds a good amount of water yet also drains well and is rich in humus - the best all-round soil for most crops. Which soil is this? [\[Soil\]](#)

- (A) Rocky soil
- (B) Pure clayey soil
- (C) Pure sandy soil
- (D) Loamy soil
- (E) Pure silt
- (F) Bedrock
- (G) Gravel

20. Work out $\frac{5}{6}$ minus $(-\frac{2}{3})$. Give the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) $-\frac{3}{2}$
- (B) $\frac{3}{2}$
- (C) $-\frac{1}{6}$
- (D) $\frac{1}{6}$
- (E) $\frac{3}{9}$
- (F) $\frac{9}{6}$ left unsimplified
- (G) $\frac{1}{2}$

21. On a sunny day a curved mirror gathers sunlight into one tiny, very bright dot that scorches a paper. Which kind of mirror is it, and where on the mirror's axis is the paper held? [\[Light\]](#)

- (A) A concave mirror, with the paper behind it
- (B) A plane mirror, with the paper at the focus
- (C) A plane mirror, with the paper very close
- (D) A concave mirror, with the paper touching the mirror
- (E) A convex mirror, with the paper at its focus
- (F) A concave mirror, with the paper at its focus
- (G) A convex mirror, far from the mirror

22. A loan of Rs 6000 is taken at 9% per year simple interest. What total amount must be repaid after 3 years? [\[Comparing Quantities\]](#)

- (A) Rs 6540
- (B) Rs 7620
- (C) Rs 7600
- (D) Rs 1820
- (E) Rs 8000
- (F) Rs 1620
- (G) Rs 7020

23. Which soil, because it holds the most water and drains very slowly, is best suited for growing paddy (rice)? [\[Soil\]](#)

- (A) Pure silt
- (B) Gravelly soil
- (C) Loamy soil that drains fast
- (D) Clayey soil
- (E) Bedrock
- (F) Sandy soil
- (G) Rocky soil

24. Write the rational number $-\frac{18}{48}$ in its standard (lowest) form. [\[Rational Numbers\]](#)

- (A) $-\frac{3}{-8}$
- (B) $\frac{3}{8}$
- (C) $-\frac{2}{8}$
- (D) $-\frac{6}{16}$
- (E) $-\frac{18}{48}$
- (F) $-\frac{3}{8}$
- (G) $-\frac{9}{24}$

25. A car's side mirror always shows a small, upright picture of traffic and gives a very wide view of the road behind. Name the mirror and describe the image fully. [\[Light\]](#)

- (A) Convex mirror; real, inverted and diminished image
- (B) Convex mirror; real, erect and enlarged image
- (C) Plane mirror; virtual, erect and same-size image
- (D) Convex mirror; virtual, erect and diminished image
- (E) Concave mirror; real, inverted and enlarged image
- (F) Concave mirror; virtual, erect and enlarged image

26. A young plant was 25 cm tall and grew to 32 cm in one month. What is the percent increase in its height? [\[Comparing Quantities\]](#)

- (A) 7%
- (B) 28%
- (C) 21.875%
- (D) 25%
- (E) 14%
- (F) 32%
- (G) 22%

27. A desert gardener grows cactus plants that rot if their roots stay wet. Which soil, draining quickly so roots do not stay soaked, suits them best? [\[Soil\]](#)

- (A) Pure silt
- (B) Waterlogged loam
- (C) Bedrock
- (D) Clayey soil
- (E) Sandy soil
- (F) Paddy-field soil
- (G) Marshy soil

28. A number is needed that sits exactly midway between $\frac{1}{3}$ and $\frac{1}{2}$ on the number line. Which number is it? [\[Rational Numbers\]](#)

- (A) $\frac{5}{12}$
- (B) $\frac{1}{12}$
- (C) $\frac{5}{6}$
- (D) $\frac{1}{2}$
- (E) $\frac{3}{5}$
- (F) $\frac{1}{5}$

29. White sunlight is passed through a glass prism and spreads out into a coloured band on a wall. How many distinct colours appear, and which colour is bent the MOST? [\[Light\]](#)

- (A) Three colours, with violet bent the most
- (B) Seven colours, with green bent the most
- (C) Two colours, with violet bent the most
- (D) Seven colours, with violet bent the most
- (E) Five colours, with red bent the most
- (F) Six colours, with violet bent the most

30. After a 20% discount, a shirt is sold for Rs 480. What was its marked price before the discount?

[Comparing Quantities]

- (A) Rs 600
- (B) Rs 540
- (C) Rs 400
- (D) Rs 576
- (E) Rs 500
- (F) Rs 576
- (G) Rs 960

31. A clayey soil sample lets only 60 mL of water pass in 30 minutes. What is its percolation rate, and does this low rate match what is expected of clay? [Soil]

- (A) 0.5 mL/min, and yes it matches clay
- (B) 2 mL/min, and yes this low rate matches clay
- (C) 20 mL/min, but no it should be slower
- (D) 30 mL/min, and yes it matches clay
- (E) 1800 mL/min, and yes it matches clay
- (F) 2 mL/min, but no, clay should drain fast
- (G) 6 mL/min, and yes it matches clay

32. Divide $(-4/9)$ by $(2/3)$ and give the answer in lowest terms. [Rational Numbers]

- (A) $8/27$
- (B) $-2/3$
- (C) $-3/2$
- (D) $-4/6$ left unsimplified
- (E) $2/3$
- (F) $-8/27$
- (G) $-6/9$

33. In the band of colours from a prism, written in order as VIBGYOR, which single colour lies exactly between blue and yellow? [Light]

- (A) Indigo
- (B) Orange
- (C) Red
- (D) Blue-green
- (E) Violet
- (F) Green
- (G) Yellow-orange

34. In a building mix, sand and cement are taken in the ratio 3 : 2. What percent of the whole mix is cement? [Comparing Quantities]

- (A) 66.67%
- (B) 40%
- (C) 30%
- (D) 60%
- (E) 20%
- (F) 50%
- (G) 25%

- 35.** The dark, crumbly material in the top layer of soil, formed from rotted dead leaves, plants and animals, is called what? [\[Soil\]](#)
- (A) Silt
 - (B) Subsoil
 - (C) Humus
 - (D) Sand
 - (E) Gravel
 - (F) Clay
- 36.** What is the additive inverse of the rational number $-5/11$? [\[Rational Numbers\]](#)
- (A) $-11/5$
 - (B) $5/11$
 - (C) 1
 - (D) 0
 - (E) $-5/11$
 - (F) $11/5$
 - (G) $-1/5$
- 37.** Two plane mirrors are fixed exactly parallel, facing each other, with a small toy between them. How many images of the toy are formed? [\[Light\]](#)
- (A) Exactly 3
 - (B) None at all
 - (C) An unlimited (infinite) number
 - (D) Exactly 360
 - (E) Exactly 4
 - (F) Exactly 1
 - (G) Only 2
- 38.** A price of Rs 2000 is first cut by 10%, and then the new price is cut by 10% again. What is the final price? [\[Comparing Quantities\]](#)
- (A) Rs 400
 - (B) Rs 1580
 - (C) Rs 1800
 - (D) Rs 1620
 - (E) Rs 1700
 - (F) Rs 1640
 - (G) Rs 1600
- 39.** Big rocks slowly break into smaller and smaller pieces over very long periods of time, helping to form soil. What is this slow breaking-down of rock called? [\[Soil\]](#)
- (A) Percolation
 - (B) Condensation
 - (C) Filtration
 - (D) Weathering
 - (E) Evaporation
 - (F) Germination
 - (G) Reflection

40. Which one of these rational numbers does NOT lie between -1 and 0? [\[Rational Numbers\]](#)

- (A) $-\frac{3}{2}$
- (B) $-\frac{7}{8}$
- (C) $-\frac{3}{4}$
- (D) $-\frac{1}{4}$
- (E) $-\frac{1}{2}$
- (F) $-\frac{1}{10}$
- (G) -0.6

41. A pen is placed between two mirrors that meet at a right angle (90 degrees). How many images of the pen are seen? [\[Light\]](#)

- (A) 2
- (B) 6
- (C) 1
- (D) infinite
- (E) 90
- (F) 3
- (G) 4

42. A shopkeeper buys 12 pens for Rs 96 and sells each pen for Rs 10. What is the profit percent on each pen? [\[Comparing Quantities\]](#)

- (A) 30%
- (B) 12%
- (C) 20%
- (D) 2%
- (E) 25%
- (F) 24%

43. In a soil profile, which layer is darkest and contains the MOST humus, making it the richest for plant growth? [\[Soil\]](#)

- (A) The subsoil (B-horizon)
- (B) The bedrock
- (C) All layers equally
- (D) The topsoil (A-horizon)
- (E) The deepest layer
- (F) The weathered-rock layer (C-horizon)

44. Arrange these rational numbers in ascending order (smallest first): $-\frac{1}{2}$, $-\frac{2}{3}$, $-\frac{3}{4}$. [\[Rational Numbers\]](#)

- (A) $-\frac{3}{4}$, $-\frac{2}{3}$, $-\frac{1}{2}$
- (B) $-\frac{2}{3}$, $-\frac{1}{2}$, $-\frac{3}{4}$
- (C) $-\frac{2}{3}$, $-\frac{3}{4}$, $-\frac{1}{2}$
- (D) they are already in order
- (E) $-\frac{1}{2}$, $-\frac{2}{3}$, $-\frac{3}{4}$
- (F) $-\frac{1}{2}$, $-\frac{3}{4}$, $-\frac{2}{3}$
- (G) $-\frac{3}{4}$, $-\frac{1}{2}$, $-\frac{2}{3}$

45. A teacher wants to throw a clear, sharp image of a glowing lamp onto a white screen. Which one of these can do it: a plane mirror, a convex mirror, or a concave mirror? [\[Light\]](#)
- (A) A concave mirror
 - (B) A convex mirror
 - (C) A plane mirror
 - (D) Only a convex mirror
 - (E) None of them
 - (F) A plane or convex mirror
 - (G) Any of the three
46. A student scored 360 marks out of 450. What percent did she score, and did she cross the 75% pass mark? [\[Comparing Quantities\]](#)
- (A) 75%, just passed
 - (B) 90%, yes she passed
 - (C) 72%, no she failed
 - (D) 80%, yes she passed
 - (E) 80%, no she failed
 - (F) 78%, yes she passed
 - (G) 80%, exactly at pass mark
47. Earthworms moving through soil make many small tunnels. How does this mainly help the soil? [\[Soil\]](#)
- (A) It cools the soil so plants die
 - (B) It stops water from entering the soil
 - (C) It makes the soil drain too fast to use
 - (D) It removes all the humus from the soil
 - (E) It lets air and water reach deeper, improving the soil
 - (F) It blocks the roots of plants
48. Find the product $(-1/2) \times (2/3) \times (-3/4)$ and give it in lowest terms. [\[Rational Numbers\]](#)
- (A) $-1/4$
 - (B) $6/24$ left unsimplified
 - (C) $3/4$
 - (D) $-6/24$
 - (E) $-1/9$
 - (F) $1/4$
 - (G) $1/9$
49. A girl 1.6 m tall stands close to a large plane mirror. How tall is her image, and what happens to the image's height if she steps back 1 metre? [\[Light\]](#)
- (A) 1.6 m, and it stays 1.6 m
 - (B) 0.8 m, and it stays 0.8 m
 - (C) 1 m, and it stays 1 m
 - (D) 1.6 m, and it grows to 2.6 m
 - (E) 1.6 m, and it shrinks to 0.6 m
 - (F) 0.8 m, and it shrinks
 - (G) 3.2 m, and it grows

50. An item bought for Rs 500 is sold for Rs 425. What is the loss percent? [\[Comparing Quantities\]](#)

- (A) 20%
- (B) 25%
- (C) 15%
- (D) 17.6%
- (E) 18%
- (F) 75%

51. In a percolation experiment, three trials take 20, 24 and 28 minutes for 200 mL of water to pass through a soil. Using the average time, what is the percolation rate (to two decimals)? [\[Soil\]](#)

- (A) About 200 mL/min
- (B) About 9.6 mL/min
- (C) About 72 mL/min
- (D) About 0.12 mL/min
- (E) About 8.33 mL/min
- (F) About 24 mL/min
- (G) About 8 mL/min

52. How many rational numbers lie between 0 and 1 on the number line? [\[Rational Numbers\]](#)

- (A) Infinitely many
- (B) Exactly 2
- (C) Exactly 10
- (D) Exactly 1
- (E) Exactly 100
- (F) None

53. Two plane mirrors between them show 8 images of a small bead. What is the angle between the mirrors? [\[Light\]](#)

- (A) 360 degrees
- (B) 8 degrees
- (C) 45 degrees
- (D) 72 degrees
- (E) 60 degrees
- (F) 9 degrees
- (G) 40 degrees

54. A fruit seller had 80 apples; 15% of them rotted and were thrown away. He sold every remaining apple at Rs 12. How much money did he collect? [\[Comparing Quantities\]](#)

- (A) Rs 144
- (B) Rs 816
- (C) Rs 768
- (D) Rs 960
- (E) Rs 900
- (F) Rs 840
- (G) Rs 720

55. Of sandy, loamy and clayey soils, which one has the GREATEST water-holding capacity? [\[Soil\]](#)

- (A) Pure silt
- (B) Gravel
- (C) Clayey soil
- (D) All hold water equally
- (E) Loamy soil
- (F) Rocky soil

56. Add $(-7/10)$ and $(3/5)$. Give the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) $-1/10$
- (B) $-10/15$
- (C) $-13/10$
- (D) $1/10$
- (E) $-4/5$
- (F) $-4/15$
- (G) $4/5$

57. Held up to a plane mirror, a printed word appears as 'NOITOM'. What word is actually written on the card? [\[Light\]](#)

- (A) MOTION
- (B) NOTION
- (C) MONITO
- (D) NOITOM
- (E) MOTOIN
- (F) NOITON
- (G) MOITON

58. Three numbers are in the ratio 2 : 3 : 5 and together add up to 400. What is the largest of the three numbers? [\[Comparing Quantities\]](#)

- (A) 160
- (B) 80
- (C) 200
- (D) 120
- (E) 240
- (F) 250
- (G) 100

59. Dumping plastic bags, broken glass and excess chemical fertilisers on land harms the ground. What is this harm to the land called? [\[Soil\]](#)

- (A) Soil weathering
- (B) Soil percolation
- (C) Air cooling
- (D) Water cycle
- (E) Soil aeration
- (F) Soil pollution

60. By how much is $\frac{3}{4}$ greater than $-\frac{1}{2}$? (Find $\frac{3}{4}$ minus $(-\frac{1}{2})$.) [\[Rational Numbers\]](#)

- (A) $\frac{1}{4}$
- (B) $-\frac{1}{4}$
- (C) $\frac{5}{4}$
- (D) $\frac{5}{2}$
- (E) $-\frac{5}{4}$
- (F) $\frac{1}{2}$

61. The word 'AMBULANCE' is painted left-right reversed on the front of the vehicle. Why is it done this way? [\[Light\]](#)

- (A) So people walking on the footpath can read it
- (B) Because paint dries better when reversed
- (C) Because mirrors cannot show normal writing
- (D) So a driver ahead sees it the right way round in their rear-view mirror
- (E) So the driver inside can read it
- (F) To save space on the bonnet

62. Out of 250 students, 12% were absent on Monday and 8% were absent on Tuesday. How many MORE students were present on Tuesday than on Monday? [\[Comparing Quantities\]](#)

- (A) 4
- (B) 40
- (C) 50
- (D) 16
- (E) 10
- (F) 20
- (G) 30

63. A child rubs two soil samples. One feels gritty and rough; the other feels smooth, fine and sticky when wet. Which is sandy and which is clayey? [\[Soil\]](#)

- (A) The gritty one is clayey; the smooth one is sandy
- (B) Both are sandy
- (C) Both are clayey
- (D) Neither can be told apart by touch
- (E) The gritty one is loam; the smooth one is silt
- (F) The gritty one is sandy; the smooth, sticky one is clayey
- (G) The smooth one is sandy; the sticky one is loam

64. Use the distributive idea to find $(\frac{2}{3}) \times (\frac{1}{2} + \frac{1}{4})$. Give the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) $\frac{2}{7}$
- (B) $\frac{1}{2}$
- (C) $\frac{3}{4}$
- (D) $\frac{2}{3}$
- (E) $\frac{5}{6}$
- (F) $\frac{1}{4}$
- (G) $\frac{1}{3}$

- 65.** A ray of light strikes a plane mirror making an angle of 25 degrees with the mirror's surface. What is the angle of reflection, measured from the normal? [\[Light\]](#)
- (A) 115 degrees
 - (B) 75 degrees
 - (C) 90 degrees
 - (D) 65 degrees
 - (E) 130 degrees
 - (F) 25 degrees
 - (G) 50 degrees
- 66.** A sum of Rs 5000 earns Rs 900 as simple interest in 3 years. What is the yearly rate of interest? [\[Comparing Quantities\]](#)
- (A) 6%
 - (B) 18%
 - (C) 12%
 - (D) 15%
 - (E) 3%
 - (F) 5%
 - (G) 9%
- 67.** Compared with the subsoil (B-horizon), the topsoil (A-horizon) is generally richer in which of these? [\[Soil\]](#)
- (A) Pure water only
 - (B) Solid unbroken rock
 - (C) Humus and living organisms
 - (D) Bedrock fragments only
 - (E) Only large stones
 - (F) Heavy metals only
 - (G) Nothing - both are identical
- 68.** Write the rational number $-\frac{5}{8}$ as an equivalent fraction with denominator 40. [\[Rational Numbers\]](#)
- (A) $-\frac{5}{8}$
 - (B) $\frac{25}{40}$
 - (C) $-\frac{10}{40}$
 - (D) $-\frac{13}{40}$
 - (E) $-\frac{5}{40}$
 - (F) $-\frac{25}{40}$
 - (G) $-\frac{40}{5}$
- 69.** A dentist uses a small mirror that shows a larger, upright picture of a tooth when held very close to it. Which type of mirror is this? [\[Light\]](#)
- (A) A convex mirror
 - (B) A flat glass sheet
 - (C) A plane mirror
 - (D) A prism
 - (E) A concave mirror
 - (F) Any curved mirror

70. A jacket marked at Rs 1500 is sold at a 12% discount. How much money does the buyer save?

[Comparing Quantities]

- (A) Rs 120
- (B) Rs 168
- (C) Rs 132
- (D) Rs 200
- (E) Rs 150
- (F) Rs 180
- (G) Rs 1320

71. Loam, the soil best for most crops, is best described as which of the following? [Soil]

- (A) Solid bedrock
- (B) Only humus with no minerals
- (C) Only broken rock with no humus
- (D) Pure clay only
- (E) Pure sand only
- (F) A balanced mixture of sand, silt, clay and humus
- (G) Pure silt only

72. What is the reciprocal of 0? [Rational Numbers]

- (A) It does not exist
- (B) 0
- (C) $1/0$ equals 0
- (D) Infinity, written as a number
- (E) -0
- (F) Any number

73. At a sharp blind bend on a hill road, a large mirror is fixed so drivers can see vehicles coming the other way over a wide area. Which mirror is best, and why? [Light]

- (A) Concave mirror, because it gives a wider view
- (B) Any mirror works equally well
- (C) Plane mirror, because it shows true size
- (D) Convex mirror, because it gives a wider field of view
- (E) Plane mirror, because it gives a wider view
- (F) Convex mirror, because it magnifies the cars

74. The price of petrol fell from Rs 100 to Rs 85 per litre. What is the percent decrease? [Comparing Quantities]

- (A) 18%
- (B) 15%
- (C) 20%
- (D) 12%
- (E) 15.5%
- (F) 17.6%
- (G) 85%

75. Two main natural factors that work over long periods to break rock and shape soil are which of the following? [\[Soil\]](#)

- (A) Only the price of land
- (B) Only plastic and glass waste
- (C) Only the colour of the rock
- (D) Only painted signs
- (E) Climate (rain, heat) and living things like plants and microbes
- (F) Only moonlight
- (G) Only electricity and magnets

76. Two rational numbers multiply together to give exactly 1. What is the special name for each number with respect to the other? [\[Rational Numbers\]](#)

- (A) They are additive inverses of each other
- (B) They are whole numbers only
- (C) They are both zero
- (D) They are reciprocals (multiplicative inverses) of each other
- (E) They are negatives of each other
- (F) They are opposite signs only

77. Which single type of mirror always forms an image that is virtual, upright, and exactly the same size as the object, whatever the distance? [\[Light\]](#)

- (A) A curved mirror
- (B) A plane mirror
- (C) No mirror can do this
- (D) A convex mirror
- (E) A concave mirror
- (F) A shaving mirror
- (G) A magnifying mirror

78. 35% of a certain number is 84. What is that number? [\[Comparing Quantities\]](#)

- (A) 294
- (B) 210
- (C) 29.4
- (D) 240
- (E) 120
- (F) 168
- (G) 250

79. On a bare hillside, heavy rain quickly washes away the soil, but where trees and grasses grow thickly the soil stays in place. What mainly holds the soil and prevents this washing away? [\[Soil\]](#)

- (A) Wind blowing over the soil
- (B) Sunlight falling on the soil
- (C) The smell of the flowers
- (D) The colour of the leaves
- (E) The height of the trees alone
- (F) The roots of plants binding the soil together
- (G) The number of branches

80. Simplify $(-2/3) - (-5/6) + (1/2)$. Give the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) $4/6$ left unsimplified
- (B) $1/3$
- (C) $5/6$
- (D) $1/2$
- (E) $-2/3$
- (F) $2/3$
- (G) $-1/3$

81. Rays of sunlight, which arrive almost parallel, fall on a concave mirror. After reflecting, where do these rays go? [\[Light\]](#)

- (A) They meet at the centre of the mirror
- (B) They meet (converge) at the focus
- (C) They spread out as if from the focus
- (D) They vanish into the mirror
- (E) They meet behind the mirror
- (F) They go back exactly the way they came
- (G) They stay parallel after reflecting

82. A trader sells goods for Rs 2400 and makes a profit of 25% on his cost. What was the cost price of the goods? [\[Comparing Quantities\]](#)

- (A) Rs 2000
- (B) Rs 1920
- (C) Rs 1900
- (D) Rs 3000
- (E) Rs 2100
- (F) Rs 1800

83. A soil sample lets 0.5 litres of water pass through it in 20 minutes. What is its percolation rate in millilitres per minute? [\[Soil\]](#)

- (A) 40 mL/min
- (B) 25 mL/min
- (C) 250 mL/min
- (D) 10 mL/min
- (E) 0.025 mL/min
- (F) 2.5 mL/min

84. The rational number $3/(-7)$ can be written more neatly with a positive denominator as which of the following? [\[Rational Numbers\]](#)

- (A) $-1/7$
- (B) $7/-3$
- (C) $3/7$
- (D) $3/-7$ is already simplest
- (E) $-7/3$
- (F) $-3/-7$
- (G) $-3/7$

85. Two mirrors are arranged so that exactly 7 images of an object appear between them. What is the angle between the two mirrors? [\[Light\]](#)

- (A) 7 degrees
- (B) 51 degrees
- (C) 40 degrees
- (D) 60 degrees
- (E) 45 degrees
- (F) 36 degrees
- (G) 90 degrees

86. A 600 mL fruit drink is 30% pure juice and the rest water. How much extra pure water must be poured in so that the juice becomes only 20% of the whole drink? [\[Comparing Quantities\]](#)

- (A) 200 mL
- (B) 120 mL
- (C) 240 mL
- (D) 180 mL
- (E) 100 mL
- (F) 300 mL

87. A potter needs soil that is smooth, sticky when wet, and can be shaped and then hardened into pots. Which soil is most suitable? [\[Soil\]](#)

- (A) Rocky soil
- (B) Gravel
- (C) Pure silt that will not bind
- (D) Clayey soil
- (E) Loose loam that crumbles
- (F) Sandy soil
- (G) Bedrock

88. Which rational number lies exactly halfway between $-\frac{1}{4}$ and $\frac{3}{4}$? [\[Rational Numbers\]](#)

- (A) $\frac{1}{8}$
- (B) $\frac{1}{2}$
- (C) $-\frac{1}{8}$
- (D) $\frac{1}{4}$
- (E) 0
- (F) $-\frac{1}{2}$
- (G) $\frac{3}{8}$

89. You stand before a plane mirror and raise your RIGHT hand. As you look at your reflection, which hand does the image appear to raise? [\[Light\]](#)

- (A) Neither hand
- (B) The image cannot raise a hand
- (C) The same hand, the right
- (D) The left hand (of the image)
- (E) It depends on the distance
- (F) Both hands
- (G) The right hand

90. A restaurant meal costs Rs 640 before a 5% service tax is added. What is the total bill? [\[Comparing](#)

[Quantities\]](#)

- (A) Rs 665
- (B) Rs 688
- (C) Rs 672
- (D) Rs 608
- (E) Rs 660
- (F) Rs 645

91. Which layer lies at the very BOTTOM of a soil profile and is the hardest, made of solid unbroken rock?

[\[Soil\]](#)

- (A) Bedrock
- (B) The A-horizon
- (C) Humus layer
- (D) The weathered-rock layer
- (E) The surface layer
- (F) Topsoil

92. Express the decimal -0.6 as a rational number in its lowest terms. [\[Rational Numbers\]](#)

- (A) $-6/5$
- (B) $3/5$
- (C) $-3/5$
- (D) $-6/10$
- (E) -0.6
- (F) $-5/3$
- (G) $-3/-5$

93. When white light is split by a prism, which colour is bent the LEAST and so appears at one end of the coloured band? [\[Light\]](#)

- (A) Yellow
- (B) Orange
- (C) Green
- (D) Blue
- (E) Red
- (F) Violet

94. A phone battery is 80% charged. During use it loses 25% OF its current charge. What percent of the full battery is left? [\[Comparing Quantities\]](#)

- (A) 50%
- (B) 65%
- (C) 75%
- (D) 55%
- (E) 20%
- (F) 60%
- (G) 45%

95. Why must a gardener water plants growing in pure sandy soil much more often than plants in clayey soil? [\[Soil\]](#)
- (A) Because sandy soil drains fast and holds very little water
 - (B) Because clay needs no water at all
 - (C) Because sandy soil holds too much water
 - (D) Because sand is always frozen
 - (E) Because sand never lets water in
 - (F) Because sand turns to rock when watered
96. Compute $(4/5)$ divided by $(-2/3)$, and give the answer in lowest terms. [\[Rational Numbers\]](#)
- (A) $-6/5$
 - (B) $8/15$
 - (C) $-12/10$ left unsimplified
 - (D) $-8/15$
 - (E) $-5/6$
 - (F) $-2/3$
97. A rainbow appears in the sky after rain when sunlight passes through tiny floating water drops. Each water drop acts like a tiny version of which optical device? [\[Light\]](#)
- (A) A concave mirror
 - (B) A convex mirror
 - (C) A magnifying glass that only enlarges
 - (D) A torch
 - (E) A black screen
 - (F) A prism
 - (G) A plane mirror
98. Which earns more simple interest in 2 years: Rs 4000 at 8% per year or Rs 5000 at 6% per year, and by how much? [\[Comparing Quantities\]](#)
- (A) The second, by Rs 40
 - (B) The first, by Rs 40
 - (C) The second, by Rs 60
 - (D) The second, by Rs 80
 - (E) The first, by Rs 80
 - (F) They are equal
 - (G) The first, by Rs 100
99. A soil has a steady percolation rate of 12 mL per minute. At this rate, how long will it take for 180 mL of water to pass through it? [\[Soil\]](#)
- (A) 6 minutes
 - (B) 2160 minutes
 - (C) 15 minutes
 - (D) 12 minutes
 - (E) 18 minutes
 - (F) 30 minutes
 - (G) 0.067 minutes

100. What number must be ADDED to $-\frac{3}{7}$ to get the result 1? [\[Rational Numbers\]](#)

- (A) $-\frac{10}{7}$
- (B) $\frac{7}{10}$
- (C) $\frac{10}{7}$
- (D) $\frac{4}{7}$
- (E) $\frac{3}{7}$
- (F) $\frac{1}{7}$
- (G) $-\frac{4}{7}$